

August 4, 2026

Alberta Machine Intelligence Institute (Amii)

Dear Dave Staszak and Hiring Team,

I am writing to apply for the Machine Learning Scientist (12 month term) position with Amii. I am a Master of Science in Computer Science graduate from the University of Calgary with hands-on experience in Python, PyTorch, applied machine learning research, model evaluation, data-processing pipelines, technical writing, and mentoring. I am excited by Amii's work at the intersection of industry and academia because it matches the kind of applied research environment where I do my best work: rigorous enough to be scientifically grounded, and practical enough to produce useful outcomes for partners.

My graduate research has focused on building reliable machine learning and data-management workflows for large, complex datasets. In my DEM super-resolution work, I trained and evaluated PyTorch ResNet models, built data collection and preprocessing pipelines with Streamlit, Rasterio, and Google Earth Engine, and assessed model behavior using RMSE, slope, TRI, TPI, and wavelet-based metrics. This work required exploratory analysis, careful validation design, reproducible experimentation, and clear communication of model tradeoffs.

I also developed a DGGS-based framework for real-time multiresolution management of spatiotemporal Earth observation data through my BigGeo/Mitacs research collaboration. The project strengthened skills that align closely with Amii's client-facing applied ML work: translating research ideas into working software, organizing data from multiple sources, evaluating technical alternatives, documenting design decisions, and explaining results to collaborators with different technical backgrounds.

The professional accomplishment I am most proud of is my first-author publication in Remote Sensing on real-time multiresolution management of spatiotemporal Earth observation data using DGGS. I am proud of it because it represents several years of independent research, engineering, experimentation, writing, revision, and collaboration brought to a peer-reviewed result. It also reflects how I like to work: deeply understand the problem, build the system carefully, measure performance, and communicate the result clearly.

Beyond research, I bring mentoring and technical communication experience. As a deep learning teaching assistant with Neuromatch Academy, I supported students through PyTorch tutorials covering MLPs, CNNs, attention and transformers, diffusion models, and reinforcement learning. As a teaching assistant at the University of Calgary, I helped students with AWS-based data workflows, large-scale data processing, SQL, debugging, and project implementation. These roles strengthened my ability to guide others through technical uncertainty while staying collaborative and practical.

I would bring Amii a strong foundation in applied ML, Python engineering, model evaluation, scientific communication, and responsible experimentation. I would welcome the opportunity to contribute to partner projects, support junior team members, and help translate machine learning research into useful industry solutions.

Thank you for considering my application. I look forward to the opportunity to discuss how my background can support Amii's Advanced Technology team.

Sincerely,

Amir Mirzai Golpayegani
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